

Exploring Graph Connectivity Through the Adjacency Matrix in Spectral Graph Theory

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Introduction to Spectral Graph Theory

Spectral graph theory is a branch of mathematics that studies the properties of a graph based on the eigenvalues and eigenvectors of matrices associated with the graph.

Contents Covered throughout the semester

- Basic definitions and types of graphs
- Types of matrices and their relationships
- Spectral theorem and quadratic forms
- Positive semidefinite matrices and Rayleigh quotient
- Courant-Fisher theorem
- Hall's spectral graph drawing and Expander graphs
- Graph coloring and Hoffman's bound for independence number
- Graph cuts and spectral clustering

Basic Graph Terminology and Definition

- **Vertices (Nodes):** Represent the fundamental units in a graph, denoted as $V = \{v_1, v_2, \dots, v_n\}$.
- **Edges (Links):** Connect pairs of vertices, denoted as $E = \{(v_i, v_j) | v_i, v_j \in V\}$.
- **Directed Graphs:**
 - Edges have a direction, denoted as $(v_i \rightarrow v_j)$.
- **Weighted Graphs:**
 - Edges are assigned weights, often representing costs or capacities.

We will focus on simple, unweighted, and undirected graphs.

Adjacency Matrix

For a graph $G = (V, E)$ with vertices $V = \{v_1, v_2, \dots, v_7\}$, the adjacency matrix A is defined as

$$A_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \in E \\ 0 & \text{otherwise} \end{cases}$$

Example: Consider the graph G with vertices $V = \{1, 2, \dots, 7\}$ and the following adjacency matrix A :

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

Eigenvalues and Eigenvectors

- **Definitions:**

- **Eigenvector:** A nonzero vector v such that when multiplied by matrix A , yields a scalar multiple of itself, i.e., $Av = \lambda v$.
- **Eigenvalue:** The scalar λ associated with the eigenvector v in the equation $Av = \lambda v$.

Solving $\det(\mathbf{A} - \lambda I) = 0$:

Eigenvalues: -2, -1, -1, 0, 0, 2, 2

k -regular Graphs and Connected Components

Definition

A graph is k -regular if every vertex has exactly k neighbors.

Definition

A connected component is a set of vertices in a graph that are connected to each other.

- By these definitions, if a graph G is k -regular, then each of its connected components X_i is also k -regular.

Theorem

Theorem

Let G be a k -regular graph with connected components $X_1, X_2, \dots, X_n$. Then, the largest eigenvalue of its adjacency matrix, $\lambda = k$, has a multiplicity equal to the number of its connected components, n .

Proof (Part 1)

Proof.

Consider a k -regular graph G composed of n connected components, denoted as $X_1, X_2, \dots, X_n$.

Define the vector $\mathbf{v}_i \in \mathbb{R}^{|V(G)|}$ for each component X_i such that

$$\mathbf{v}_i(j) = \begin{cases} 1 & \text{if vertex } j \text{ belongs to } X_i, \\ 0 & \text{otherwise.} \end{cases}$$

Given the k -regularity of X_i , the adjacency matrix A_i of X_i has the property that every row sum is k . Therefore, the product $A_i \mathbf{v}_i$ results in each entry of the resultant vector being k , confirming $\mathbf{v}_i$ as an eigenvector of A_i associated with the eigenvalue k . □

Continued.

By ordering vertices such that those belonging to the same X_i are contiguous, A can be expressed as a block-diagonal matrix:

$$A = \begin{bmatrix} A_1 & 0 & \cdots & 0 \\ 0 & A_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & A_n \end{bmatrix}.$$

This block structure ensures that the eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ corresponding to k are orthogonal and linearly independent, each localized to its respective component. Hence, the eigenvalue k has multiplicity at least n . □

Proof (Part 3)

Continued.

Let λ be an eigenvalue of A with a corresponding eigenvector $\mathbf{x}$. Normalize $\mathbf{x}$ such that the largest component in magnitude is 1, i.e., let $|x_j| = 1$ for some j and $|x_i| \leq 1$ for all other i .

Then $A\mathbf{x} = \lambda\mathbf{x}$ for the j -th component simplifies to:

$$(Ax)_j = \sum_{i=1}^n A_{ji}x_i = \sum_{i \text{ adjacent to } j} x_i = \lambda x_j$$

Applying the triangle inequality:

$$|\lambda| = |\lambda x_j| = \left| \sum_{i \text{ adjacent to } j} x_i \right| \leq \sum_{i \text{ adjacent to } j} |x_i| \leq k$$



Conclusion of Proof

Concluding Part.

Since A is block-diagonal with each block A_i contributing exactly one independent eigenvector corresponding to k , and knowing that k is the upper bound for any eigenvalue of A , it follows that the multiplicity of k is exactly n . There are no other vectors spanning across components due to the zero blocks in A , ensuring no additional eigenvectors for k . Thus, the eigenvalue k has multiplicity exactly n , equal to the number of connected components in the graph. □

- **Real-world Applications:**

- *Network Analysis*: Analyzing social networks, communication networks, and transportation networks.
- *Chemistry*: Understanding molecular structures, predicting chemical properties, and designing molecular networks.
- *Physics*: Quantum networks, quantum walks

Thank You!